1. GENERATE FOUR SETS OF SIMULATION WITH TWO GEOMETRIES
(example with the raw configuration)

FISH DIGITAL TWIN

Constraints included:

- Automatic time resolved kinematics
- Body deformation
- 256 points for realistic body shape

$X_{fish}(t)$



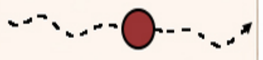
Simulated pressure $p_{fish}(t)$ at sensor locations

CYLINDER SURROGATE

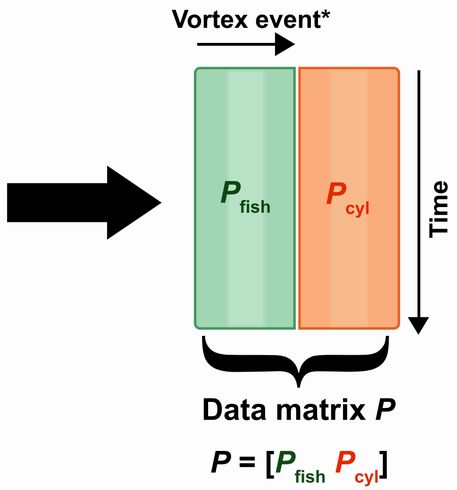
Constraints included:

- Same center-of-mass trajectory as the fish
- Rigid body (no deformation)
- 2D circular cross-section

$X_{cyl}(t)$



Simulated pressure $p_{cyl}(t)$ at sensor locations

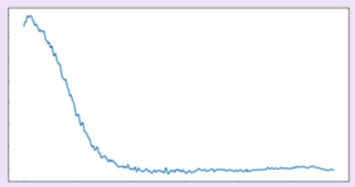


*A vortex event correspond to one or multiple vortices passing in front of a sensor without time discountinuity

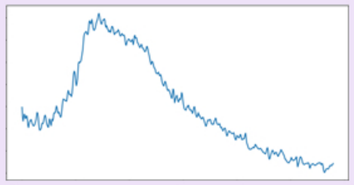
2. COMPUTE SVD AND TRAIN LINEAR CLASSIFIER (example with two SVD features)

A. Compute SVD

$P = U\Sigma V^T$



SVD mode U_1



SVD mode U_2

B. Project onto SVD modes

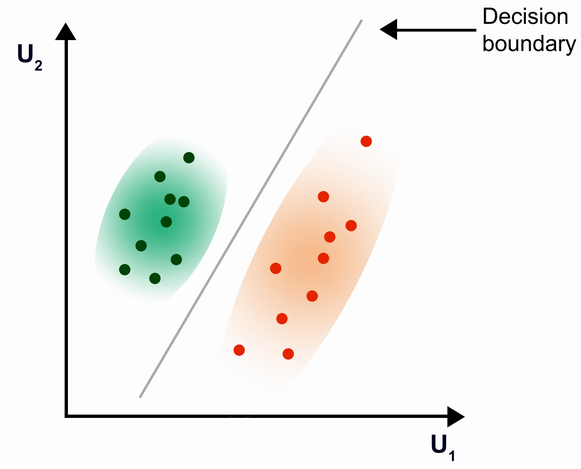
$x_{i\ fish} = P_{i\ fish} \cdot U =$

$\frac{P_{i\ fish} \text{ projected on } U_1}{P_{i\ fish} \text{ projected on } U_2}$

$x_{i\ cyl} = P_{i\ cyl} \cdot U =$

$\frac{P_{i\ cyl} \text{ projected on } U_1}{P_{i\ cyl} \text{ projected on } U_2}$

C. Train linear classifier

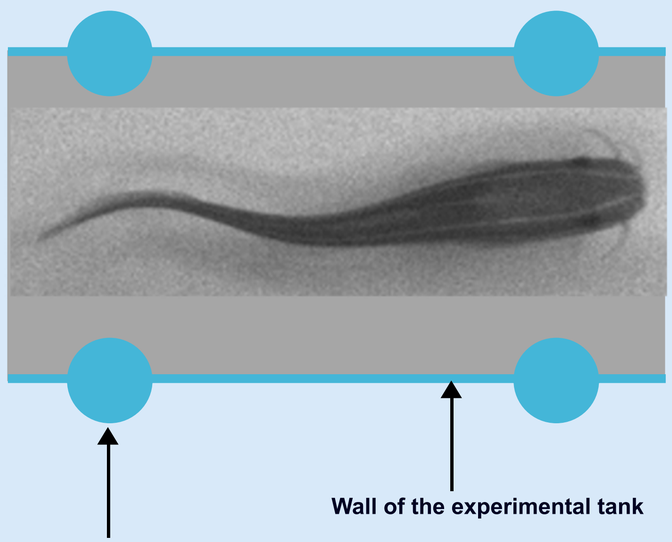


3. INFERE ON REAL MEASUREMENTS

A. Measure experimental pressure

EXPERIMENTS

- 40 freely swimming Stripped eel catfish
- Experiments during 15min



B. Project onto SVD modes

$x_{i\ meas} = P_{i\ meas} \cdot U =$

$\frac{P_{i\ meas} \text{ projected on } U_1}{P_{i\ meas} \text{ projected on } U_2}$

Project measurements into the same feature space (U is the one from step 2.)

C. Infere on the trained linear classifier

